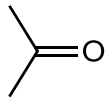
3(a)(i)		**1**

3(a)(ii)	yellow precipitate	**1**

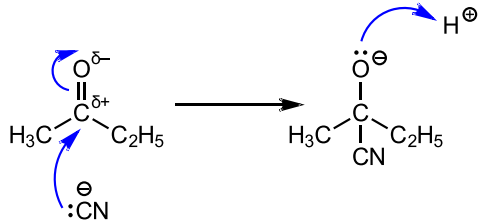
Examiner report missing
9701_w25_qp_24 Question 3

6(a)	carbonyl	**1**
6(b)	**M1** (2,4)-dinitrophenylhydrazine / (2,4)-DNPH **M2** orange / red / yellow precipitate	**2**
6(c)(i)	**F** = propanone **G** = ethanal **H** = propanal	**2**
6(c)(ii)	tri-iodomethane **OR** CHI_3	**1**

Examiner report missing
9701_s25_qp_21 Question 6

4(a)		**5**

acidified $K_2Cr_2O_7$(aq)	(solution) turns (from orange to) green	**J** and **L**
alkaline I_2(aq)	yellow precipitate forms	**K** and **L**
2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	**J** and **K**
Na(s)	effervescence	**L** and **M**

4(b)(i)		**4**

- curly arrow from lone pair on :CN to C($^{\delta+}$)
- curly arrow from C=O to O($^{\delta-}$) **AND** correct dipole
- correct intermediate
- curly arrow from lone pair on O:$^-$ to H^+

4(b)(ii)	hydroxynitrile	**1**

Examiner report missing
9701_m25_qp_22 Question 4

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(b)(ii)	ultraviolet	**1**
2(c)(iii)	CH_3COCH_3 + **3**NaC*l*O → **1**CHCl_3 + **1**CH_3COONa + **2**NaOH	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

(iv) Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

(b) (i) Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

(ii) This question was well answered.

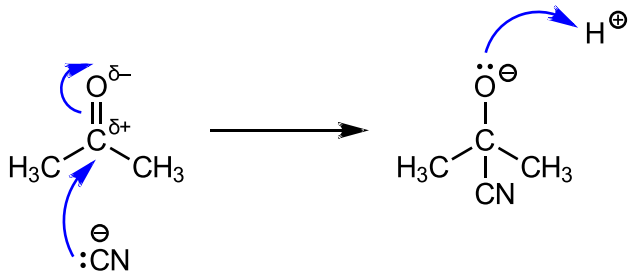
(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

(c) (i) Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(c)(i)	CH_3COCH_3 + **2**[H] → $CH_3CH(OH)CH_3$	**1**

4(d)	red / orange / yellow ppt	**1**

4(e)	Fehling's solution cannot **oxidise** ketones OR ketones are not easily **oxidised**	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

(b) Candidates often stated that the mineral acids HCl or H_2SO_4 were required for the hydrolysis reaction of a CN group to a carboxylic acid but missed the requirement for the acid to be dilute or aqueous.

(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

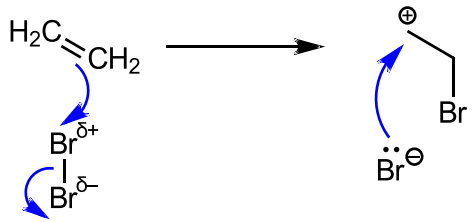
(ii) The name of compound C was well known.

(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**

4(d)(i)	structural / positional	**1**

4(d)(iii)		**3**

reagent	observation with **F**
2,4-DNPH	red / orange / yellow ppt
Tollens' reagent	silver mirror OR grey ppt OR black ppt
alkaline I_2(aq)	yellow ppt

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(b)(ii)	ultraviolet	**1**
2(c)(iii)	CH_3COCH_3 + **3**NaC*l*O → **1**CHCl_3 + **1**CH_3COONa + **2**NaOH	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

(iv) Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

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(ii) This question was well answered.

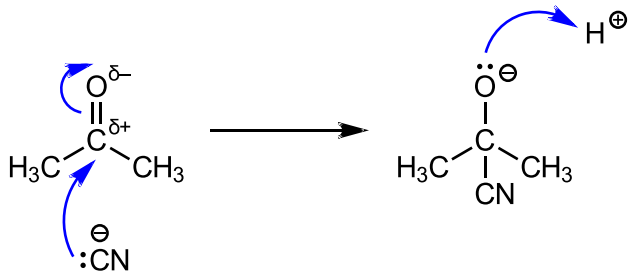
(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

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(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(c)(i)	CH_3COCH_3 + **2**[H] → $CH_3CH(OH)CH_3$	**1**

4(d)	red / orange / yellow ppt	**1**

4(e)	Fehling's solution cannot **oxidise** ketones OR ketones are not easily **oxidised**	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

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(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

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(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

Question 5

(a) (i) Those candidates with a good understanding of mass spectrometry realised that the peaks shown in Table 5.1 represented the molecular ion peak, M^+, and its $(M+1)^+$ peak and used the appropriate formula to calculate the relative abundance of the M+1 peak. Some weaker responses based their calculation on an incorrect formula.

(ii) The strongest responses engaged with the information about compound **W**. Weaker responses did not engage fully with the question details and answered in terms of structural formulae. Some formulae gave the fragments with no charges or negative charges.

(b) (i) An understanding that an O–H bond, due to a carboxyl functional group, produces a broad stretch within the range 2500–3000 cm^{-1} was seen in the best responses. The weakest responses did not engage with the details given for compound **W** and its reaction to produce **X**, and listed every potential functional group which might account for bond **A** and bond **B**, including amide and ester functional groups.

(ii) Of those candidates that correctly identified **B** as a C=O of a carbonyl group, some candidates ignored the reaction conditions stated in the question and represented **X** as an aldehyde rather than a ketone.

(c) (i) A significant proportion of candidates correctly named the functional group present in **Y**, based on the information provided.

(ii) Many equations were shown with the correct formula of the organic product; not all equations were balanced. Identification of H_2O rather than H_2 as the other product was seen.

(iii) Those candidates who applied knowledge of the behaviour of different types of alcohol and carbonyl compounds correctly deduced the structure of **Y**. Confusion about the chemical reactions of alcohols and carbonyl compounds was seen in weaker responses.

6(a)(i)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	**1**

6(c)(i)	**M1** deduce n = 12 (from **Y** – 1C (in methanol)) **M2** $(100 \times x) \div (1.1 \times 100) = 12$ (so) $x = 13.2$	**2**

6(c)(ii)	<table><tr><th></th><th>observation on addition of 2,4–DNPH</th></tr><tr><td>**Y**</td><td>orange precipitate</td></tr><tr><td>**Q**</td><td>no precipitate</td></tr></table> Both correct for one mark	**1**

Question 6

(a) (i) The definition of stereoisomerism was one of the least well answered questions. The most common error was to state that 'molecules had the same molecular formula but different structural formula'. Few mentioned that it was the arrangement of the atoms or groups in space.

(ii) Very few answers gained full credit here. Some candidates identified a C=C double bond and two chiral centres and then incorrectly stated that this gave (2 × 3 =) 6 stereoisomers instead of using the correct expression 2^3 (= 8).

(iii) The molecular formula of **V** was answered quite well. There were no common errors.

(iv) Many candidates correctly identified an alkene and carbonyl group. The most common error was to omit the ester group and substitute it for an alcohol group or a carboxyl/carboxylic acid group.

(b) (i) The question required an identification of the role of reagent **T**. Candidates who stated that **T** behaved as a reducing agent gained full credit. Those candidates who ventured to explain the behaviour of **T** with each functional group generally understood that it reduced the carbonyl group in **V** to an alcohol group in **W**. The reaction of the C=C to from a saturated chain attracted a variety of incorrect answers ranging from acting as a catalyst to hydrolysing the alkene group.

(ii) This question was well answered.

(c) (i) Candidates found this very challenging. Many answers used the m/e values of 200 and 201, from the mass spectrum, instead of the relative abundance values of 100 and x, and attempted to average these values to determine the relative atomic mass.

Creditworthy answers deduced that **Q** contained 12 C atoms and then used this value of *n* in the expression $n = \dfrac{100 \times \text{abundance of M} + 1 \text{ ion}}{1.1 \times \text{abundance of M} + \text{ion}}$

to calculate the abundance of the (M + 1) ion which is equivalent to the value of x.

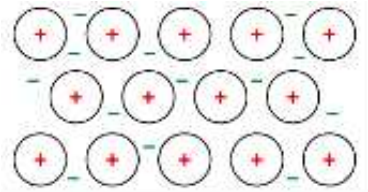
(ii) This question was answered correctly by many candidates with the most common error being to omit 'precipitate' with the orange colour for the observation with 2,4-DNPH with **Y**.

(iii) The majority of candidates did not recognise that the volume of hydrogen gas was collected at s.t.p., in the question, which, from the Table of data, is 22.4 $dm^3 mol^{-1}$.

Various methods were used to calculate the moles of gas produced to include the ideal gas equation and 24.0 $dm^3 mol^{-1}$.

Those candidates who managed to calculate the number of moles of gas then needed to appreciate that the ratio of hydroxyl groups to hydrogen gas was 2 : 1. The award of full credit was rare.

(iv) Many candidates gave correct absorption ranges for one of the two carbonyl absorptions in **Y**, for the ketone and ester functional groups. Several answers gave a range of absorptions for a carboxyl group in **Q** as a difference in the infrared spectrum, having not appreciated that **Q** contains **only** hydroxyl groups, mentioned in the stem of **(c)(ii)**.

1(a)	 labelled diagram shows: • regular arrangement of positive ions • **AND** surrounded by / (sea of) delocalised electrons	**1**

1(d)(i)	a substance that removes electrons from another / causes electron loss / gains electrons	**1**

1(d)(ii)	($KMnO_4$ is) **cold** and **dilute**	**1**

1(d)(iii)

reagent	observation with **Y**	observation with **Z**
Na_2CO_3(aq)	no reaction	● no reaction
alkaline I_2(aq)	● yellow precipitate (ppt)	● yellow ppt
2,4-DNPH	● red / orange / yellow ppt	● red / orange / yellow ppt
Tollens' reagent	● no reaction	● silver mirror

4

1(d)(iv)	$CH_3CHO + 2[H] \rightarrow CH_3CH_2OH$	**1**

Question 1

This was a broad-based question, encompassing ideas of physical, inorganic and organic chemistry. Candidates were adept with simple definitions, but application of ideas showed some deficit of understanding.

(a) Many diagrams were well presented, but a substantial number neglected to label the particles: mention of delocalisation of electrons was crucial. A sizable minority incorrectly identified the positive particles as protons or nuclei.

(b) (i) This question was mostly well answered, though errors of balancing and 'Bi_2' were often seen.

(ii) Although many candidates secured full credit here, common errors were not to link deduction and reasoning (e.g. giant *because* high melting point) and to offer contradictory statements (e.g. giant molecular and ionic bonding). In many cases, brevity served better. Some candidates appeared confused about the compound they were being asked to describe, Bi_2O_3, rather than bismuth itself.

(c) (i) This was answered well, though answers of +6 were commonly seen with respect to bismuth.

(ii) Many different answers were offered here, including $NaBH_4$ and O_2, whereas candidates needed to refer to a compound given in the equation.

(d) (i) This was generally well answered. It is preferable to refer to transfer of electrons when explaining the term oxidising agent.

(ii) Some contradictions were observed here, such as 'cold dilute … heat under reflux.'

(iii) Many candidates answered this well. Common errors included a switch of answers in the case of Tollens' reagent and neglecting to mention a precipitate or solid in the iodoform and DNPH reactions.

(iv) Many candidates tried unsuccessfully to include $NaBH_4$ in their answers. Candidates are reminded that even with the use of [H], there is a need for a stoichiometric balancing number. Use of CH_3COH is discouraged as a method of representing aldehydes.

(e) The initial stages of most calculations were good, but many candidates then forgot to multiply by 40 (to scale up to 1 dm^3 from 25 cm^3) or to multiply the concentration by 10 (1000 ÷ 100). In general, working needed to be shown more clearly, as some inventive methods could otherwise have given a seemingly correct answer.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(e)(i)	1 4 0	**1**

3(e)(ii)	Solution turns from purple to colourless	**1**
	Orange / red precipitate forms	**1**
	no reaction	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

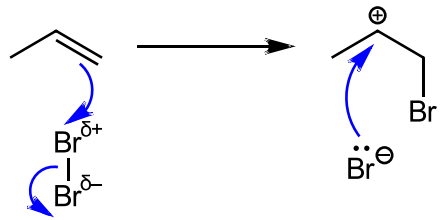
Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

4(a)(i)	(it has molecules that) contain a chiral carbon / centre **OR** are non-superimposable mirror images	**1**

4(b)(i)	 curly arrow from C=C bond to $Br^{\delta+}$ **AND** curly arrow from $Br^{\delta+}$— $Br^{\delta-}$ bond to $Br^{\delta-}$ **AND** $Br^{\delta+}$ closest to the double bond	**1**
	correct intermediate	**1**
	curly arrow from lone pair on Br^- to $C^{(+)}$ of the intermediate	**1**

4(b)(iii)	HCN&KCN **OR** NaCN with H_2SO_4	**1**
	NaOH(aq) (1) dilute / aq(ueous)HC*l* / H_2SO_4 (1)	**2**
	acidified $K_2Cr_2O_7$	**1**

Question 4

(a) (i) Some candidates gave clear explanations of the presence of a chiral carbon in the lactic acid structure. A few candidates stated that optical isomers had chiral centres or described the behaviour of an optical isomer in the presence of polarised light.

(ii) Many candidates were unable to give the systematic name of lactic acid. 'Carboxylic acid' was seen frequently.

(iii) It was common for diagrams to show **more** than one water molecule or **multiple** hydrogen bonds. Candidates should be advised to respond as directed by the question. It was quite common for answers, showing more than one hydrogen bond, to contradict a correct representation.

(b) (i) Some clear and precise reaction mechanisms were shown with partial charges assigned to relevant atoms and unambiguous representation of curly arrows to show bonds broken and bonds made. When drawing mechanisms of this type candidates should be encouraged to consider what the curly arrows represent and whether their representation is clear, use partial charges correctly and discriminately and check that the number of carbon atoms on the intermediate is correct.

(ii) The equations described sometimes contained only pyruvic acid as a product or identified H_2 rather than H_2O as a product.

(iii) This question was demanding. Many candidates confused the reagents and conditions required to convert halogenoalkanes to nitriles with those required to convert ethanal to a hydroxynitrile. Deduction of the reagents and conditions used in the second step of both synthesis from propene and ethanal proved demanding for most candidates. Oxidising agents were more commonly seen for the third steps, however, some candidates represented the oxidising agents with incorrect formulae or with no reference to acid conditions.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(e)(i)	1 4 0	**1**

3(e)(ii)	Solution turns from purple to colourless	**1**
	Orange / red precipitate forms	**1**
	no reaction	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

5(a)	C_3H_7O	**1**
5(b)(i)	$C_6H_{10}O_3 + [6H] \rightarrow C_6H_{14}O_2 + H_2O$	**1**
5(b)(ii)	$LiAlH_4$ (in dry ether)	**1**
5(d)	<table><tr><td>reagent</td><td>observation on addition to **X**</td><td>observation on addition to **Y**</td></tr><tr><td>sodium carbonate solution</td><td>effervescence</td><td>no change</td></tr><tr><td>2,4-DNPH</td><td>orange precipitate</td><td>no change</td></tr><tr><td>alkaline aqueous iodine</td><td>yellow precipitate</td><td>yellow precipitate</td></tr></table> Two correct for one mark, four correct for two marks, six correct for three marks	**3**

Question 5

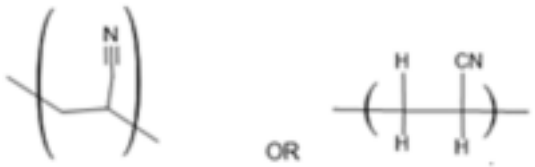
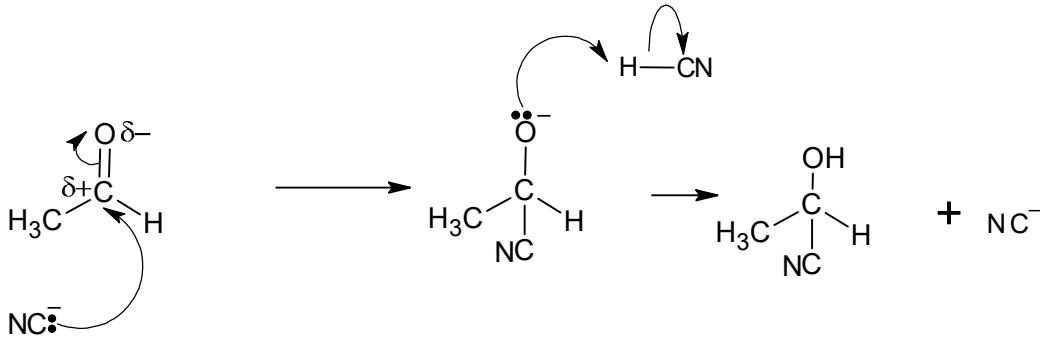
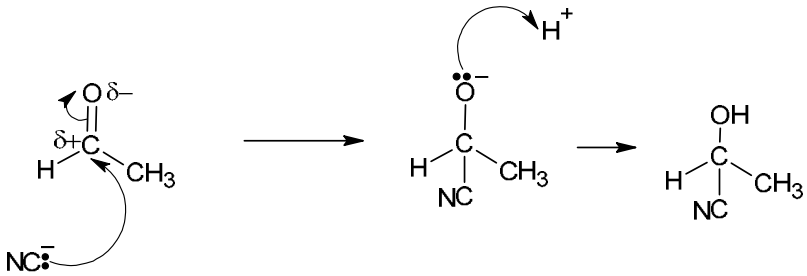
(a) Some candidates deduced the correct empirical formula of C_3H_7O from a molecular formula of $C_6H_{14}O_2$.

(b) (i) Very few candidates used molecular formula, as required in the question, to construct an equation for the reduction reaction.

(ii) Many answers correctly stated that a suitable reducing agent for the reaction is $LiAlH_4$.

(c) Some good answers were seen.

(d) Many candidates correctly stated that **X**, the ketoacid, would react with all three reagents to give effervescence, an orange precipitate and yellow precipitate respectively. Fewer candidates recognised that the diol, **Y**, would only react with the alkaline $I_2(aq)$ as it contained a $CH_3CH(OH)$ moiety.

6(a)	 OR	**1**
6(b)(i)	nucleophilic addition	**1**
6(b)(ii)	 **OR**  **M1** correct dipole on C=O **AND** curly arrow from C=O bond to O **AND** from lone pair of electrons on C of :CN^- **M2** correct intermediate (charge on O or overall) **M3** curly arrow from lone pair of electrons on O to H^+ **AND** correct organic product (+ CN^-)	**3**

Question 6

(a) The single repeat unit derived from the monomer, CH_2=CHCN, by addition polymerisation was quite well answered.

(b) (i) Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

(ii) Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the CN^- to the C (in the C=O bond) did not start from the lone pair on the C of the CN^- ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the CN^- nucleophile and many answers incorrectly showed an arrow from an H^+ ion to an O^- (in the intermediate).

(iii) Some good answers were seen.

(iv) The absorptions in the infrared spectrum of CH_2=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

(c) (i) Very few candidates gave an accurate description.

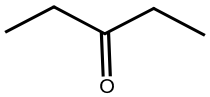
(ii) A number of answers were awarded some credit. Few candidates were able to give a full description.

(d) Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

(e) The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

(f) (i) It was well known that the name of the type of reaction in step 1 is substitution.

(ii) Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element **AND** different structural formulae / different structures	**1**
5(b)(i)	carbonyl (group)	**1**
5(b)(ii)	 **A**  **B**	**2**

Question 5

(a) This definition was generally well known.

(b) (i) Many candidates appreciated that the two molecules were members of the carbonyl functional group. Some candidates incorrectly referred to them as either aldehydes or ketones.

(ii) Correct structures of **A** were common. Many candidates represented **B** as a ketone; the correct carbon skeleton was not always deduced.

(c) (i) It was common for some of the information in the question to be used to produce partially correct structures. Many candidates incorrectly produced structures with two types of stereoisomerism.

(ii) This was generally well answered.

(d) (i) Ketone and carboxylic acid were the most common incorrect answers.

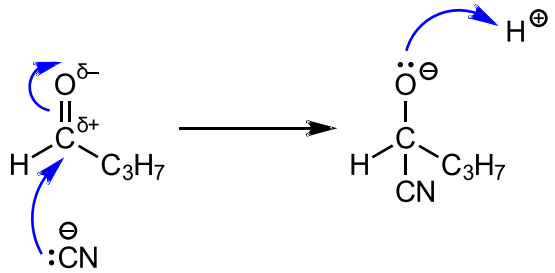
(ii) Hydrolysis was a common incorrect response.

(iii) Deduction of the structure of **D** proved to be demanding. Often the structure drawn had a carboxylic acid group at one end of the molecule; identification of $(CH_3)CH(OH)$– at one end of the molecule was seen less frequently. Occasionally, the correct ends of the molecule were drawn with an incorrect number of carbon atoms in the carbon backbone.

(iv) Stronger candidates clearly identified two differences between the spectra in terms of the different absorption ranges produced as a result of the different environment of the C=O bond in an ester compared to a carboxyl group, or in terms of the presence of the O-H bond in a hydroxyl or carboxyl group which is absent in the ester, **E**. Reference to absorbance at 2850 – 2950 due to C-H bonds was common, even though both **D** and **E** contain this type of bond.

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) **one product**	**1**

3(a)(iii)	butanal	**1**

3(b)(i)	 curly arrow from lone pair on C of CN^- to $C^{(\delta+)}$	**1**
	correct dipole on C=O **AND** curly arrow from C=O bond to $O^{\delta-}$	**1**
	correct organic intermediate	**1**
	curly arrow from lone pair on alkoxide $O^{(-)}$ to H^+ / H of HCN / H of H_2O (to reform the alcohol)	**1**

3(b)(ii)	acidified $K_2Cr_2O_7$	**1**
	Tollens' reagent	**1**
	no reaction	**1**

Question 3

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

(a) (i) Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

(ii) This was well answered by most candidates.

(iii) The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

(iv) Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

(b) (i) Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the CN^- ion.
- giving the intermediate a positive charge which was subsequently attacked by the CN^- nucleophile.
- mechanisms involving the H (of HCN) / H^+ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

(ii) The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

(iii) There were some very good answers here, although some increased the number of H atoms in the product and/or H_2 appeared as a product.

(iv) This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

(c) It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

(d) (i) Some answers referred to a change in position of the equilibrium/yield, which was not credited.

(ii) Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

(iii) Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

(iv) Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.